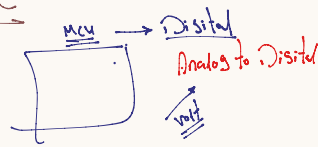
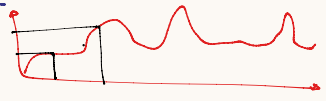


- ADC

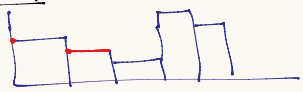


Analog

* Analog

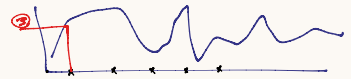


* Digital

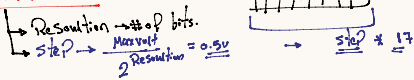


* Convert from Analog to digital

① Sample & Hold

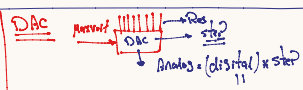


② Quantization

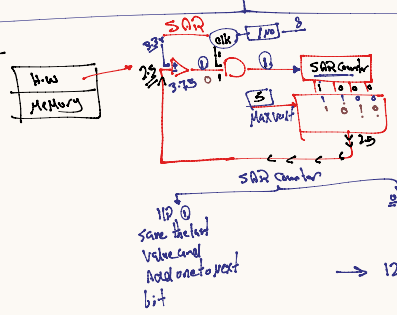


* Notes

- ① ADC → Volt Analog
 - ② ADC → Quantization
 - ③ Resolution → Fixed → How Accurate
- Max Volt → 4
Step = $\frac{4}{2^2} = 1$
= Max Volt / Resolution
- No of bit = 2
00 → 0V
01 → 1V
10 → 2V
11 → 3V



ADC Peripheral



- start
- ① SAR Counter add 1 at MSB
 - ② DAC operation
analog = digital * step
 $100 * \frac{5}{2} \Rightarrow 8 * \frac{5}{16}$
 $\frac{25}{2}$
 - ③ OP AMP Compare between SAR/DAC value
- 110 ① same the last value and add one to next bit
 $\rightarrow 12 * \frac{5}{16} \Rightarrow 3.75$

① ADC → Volt Separate

A_{VCC} G_{ND} → V_{CC}

② ADC → DAC must be work

need Volt → Reference Volt
→ Ref

Sensor type

① digital sensor

② Analog sensor

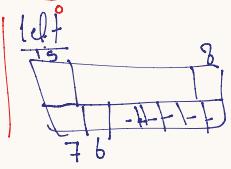
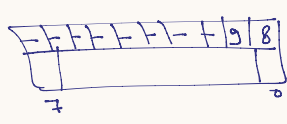


Left adjustment for ADC

* Result



Right Adjust



AD Mux

- ① Address
- ② Initial
- ③ type
- ④ each Bit what do.

* ADC

$$\frac{5}{2^{10}} = \frac{5}{1024}$$

13 → 260 μsec → Conversion time

KSPS → kilo sample per sec

* 8 single ended

